

UME FARWA RIZVI

Computer Science Student | AI, Data Science & Software Development

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PROFESSIONAL SUMMARY

University student with a strong interest in Artificial Intelligence, software development, and machine learning. Skilled in programming, problem-solving, and teamwork through academic projects. Seeking hands-on technical exposure in a growth-oriented tech environment.

EDUCATION

Bachelor of Science in Computer Science — Data Science

2023 – 2027 (Expected)

COMSATS University Islamabad, Wah Campus | 7th Semester | CGPA: 3.28 / 4.0

- Relevant Coursework: Data Structures & Algorithms, Database Design, Data Mining, OOP, AI, Machine Learning

INTERNSHIP EXPERIENCE

Systems Department Intern

Aug 2025 – Sep 2025

Oil & Gas Development Company (OGDC), Islamabad

- Completed 4-week internship gaining hands-on experience with SAP ERP GUI in a corporate enterprise environment.
- Assisted with system operations, data management, and ERP workflows; gained insight into enterprise-level business process management.

PROJECTS

BreastCare AI | Python | Streamlit | LightGBM | XGBoost | SHAP | FAISS | Gemini API

Role: AI & Data Science Developer

- Developed a multi-layered AI decision-support system integrating machine learning classifiers, SHAP explainability, and RAG pipelines for breast cancer risk estimation.
- Built a 5-layer pipeline retrieving peer-reviewed ACR BI-RADS guidelines to ground Gemini-generated explanations, achieving ~84% accuracy and ensuring compliant, non-diagnostic output via code-level safety filters.

5G Network Slicer | Python | Docker | React.js | Next.js | FastAPI

Role: Implementation of FastAPI, Microservices, Concurrency

- Designed a containerized microservices system supporting 10+ concurrent network slices and handling 100+ API requests/min with ~95% success rate.
- Implemented RESTful backend services enabling real-time network resource allocation across independent slices.

Movie Management System | PHP | MySQL

Role: Backend Developer

- Developed secure web system with user authentication, full CRUD operations, and 3NF-normalized relational database design.
- Built a secure backend system managing 100+ records across 5+ normalized tables, reducing data redundancy by ~30%.

Road Accident Severity Prediction System | Python | Scikit-learn | Streamlit

Role: Model Training and Testing

- Trained 3 ML models achieving ~85% accuracy and improved performance by 20% using SMOTE on a dataset of 10,000+ records.
- Deployed real-time prediction dashboard via Streamlit with data visualization and model performance comparison.

Weather Forecast Application | Flutter | Dart | REST API

Role: Full Stack Developer

- Built cross-platform mobile app with live REST API integration, geolocation services, and real-time weather rendering.
- Developed a cross-platform app with real-time API integration delivering <2 sec response time and dynamic updates with <1 sec UI refresh.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, Dart, SQL, PHP

Frameworks & Libraries: FastAPI, Flask, Flutter, React.js, Next.js, Streamlit, Pandas, NumPy, Scikit-learn

Tools & Platforms: Docker, MySQL, Git, VS Code, IntelliJ IDEA, NetBeans, SAP ERP, Tableau, Power BI, Excel

Core Concepts: OOP, REST APIs, Microservices, Database Design (3NF normalization), Data Mining, Data Visualization, Explainable AI (SHAP), RAG

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Content Manager – COMSATS Media Club:** Led and coordinated a 10+ member team, streamlining content production and improving publication consistency.
- CO-Media Lead – COMSATS Literary Society (CLS):** Directed a 10-12 member media team, driving event promotions and boosting audience reach through targeted content strategies.

LANGUAGES & CERTIFICATIONS

- English:** Fluent — IELTS Academic Band 7.0
- Urdu:** Native